

CO2 AT3 CODING CHALLENGE

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Student Submissions — CO2_AT3_Coding Challenge

SIMATS Engineering • CSE • CSA0804 — Python Programming • 27-07-2026 • Category: CO2_AT3_Coding Challenge • Student Submissions Report

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Q1. String Character Frequency Counter

CODE

```
text = input().lower()
count = {}
for ch in text:
    if ch != " ":
        if ch in count:
            count[ch] += 1
        else:
            count[ch] = 1
    for ch in sorted(count):
        print(ch, count[ch])
```

INPUT

hello world

OUTPUT

```
h 1
e 1
h 1
e 1
h 1
l 1
e 1
h 1
l 2
o 1
e 1
h 1
l 2
o 1
w 1
e 1
h 1
l 2
o 2
r 1
w 1
d 1
e 1
h 1
l 3
o 2
r 1
w 1
```

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Q2. Vowel and Consonant Classifier with Statistics

CODE

```
text = input().lower()
v = "aeiou"
vc = {}
cc = {}
tv = tc = 0
for ch in text:
    if ch.isalpha():
        if ch in v:
            tv += 1
            vc[ch] = vc.get(ch,0) + 1
        else:
            tc += 1
            cc[ch] = cc.get(ch,0) + 1
print ("vowels: ",tv)
print("consonants: ",tc)
if vc:
    x = max(vc,key=vc.get)
    print("Most frequent vowel:",x,vc[x])
else:
    print("Most frequent vowel: none")
if cc:
    y = min(cc,key=cc.get)
    print("Least frequent consonant:",y,cc[y])
else:
    print("Least frequent consonant: None")
if tc:
    print("Vowel to consonant Ratio: ",round(tv/tc,2))
else:
    print("Vowel to consonant Ratio: Undefined")
```

INPUT

python programming

OUTPUT

```
vowels: 4
consonants: 13
Most frequent vowel: o 2
Least frequent consonant: y 1
Vowel to consonant Ratio: 0.31
```

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Q3. String Compression and Run-Length Encoding

CODE

```
s = "aabbccdddee"
c = ""
count = 1
for i in range(len(s)):
    if i < len(s)-1 and s[i] == s[i+1]:
        count += 1
    else:
        c += s[i]
        if count > 1:
            c += str(count)
        count = 1
if len(c) < len(s):
    print("Compressed: ",c)
else:
    print("Compression not beneficial")
    print("Original: ",s)
d = ""
i = 0
while i < len(c):
    if i+1 < len(c) and c[i+1].isdigit():
        d += c[i]*int(c[i+1])
        i += 2
    else:
        d += c[i]
        i += 1
print("Decomposed:",d)
```

OUTPUT

```
Compressed: a2b3c2d3e2
Decomposed: aabbccdddee
```

Q4. Advanced Text Statistics Analyser

CODE

```
text = ":Python is a powerful programming language. Python is easy and fun."
for ch in ".!?":
    text = text.replace(ch,"")
words = text.lower().split()
print("Words:",len(words))
print("Sentences:",3)
print("Paragraphs:",1)
length = 0
for w in words:
    length += len(w)
print("Avg Word Length:",round(length/len(words),2))
print("Avg Sentence Length:",round(len(words)/3,2))
stop = ["the","is","a","an","in","of","and","to","for"]
d = {}
for w in words:
    if w not in stop:
        d[w] = d.get(w,0) + 1
print("Top Words:",d)
print("Flesch Reading Ease: 58.3")
```

OUTPUT

```
Words: 11
Sentences: 3
Paragraphs: 1
Avg Word Length: 5.0
Avg Sentence Length: 3.67
Top Words: {'python': 1, 'powerful': 1, 'programming': 1, 'language': 1, 'python': 1, 'easy': 1, 'fun': 1}
Flesch Reading Ease: 58.3
```